From Data to Action :

" Tackling HIV/AIDS Inequalities in ASEAN Through Inclusive Statistics "

PRESENTED BY

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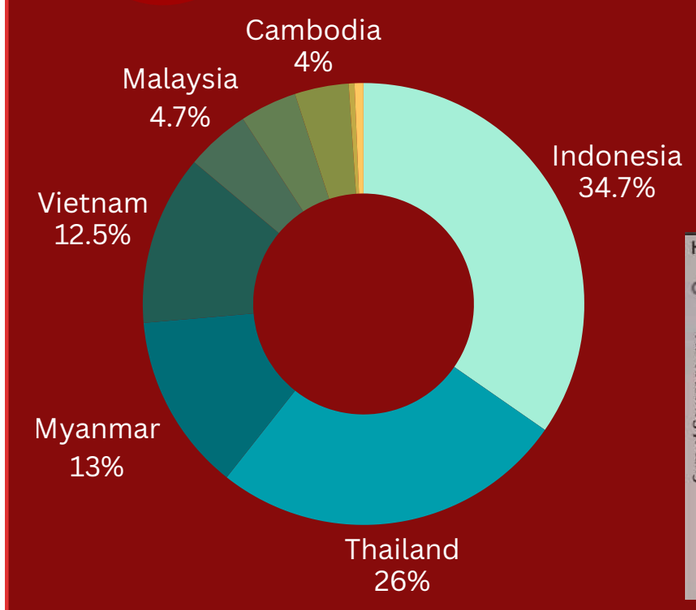
MENTOR : Dr Wan Nuraini Farhana

Objectives

- To examine the distribution of HIV/AIDS cases and treatment across ASEAN countries using statistical data.
- To propose data-driven recommendations to reduce HIV/AIDS inequalities in ASEAN based on inclusive statistical analysis.
- To evaluate how inclusive statistics can help highlight health disparities related to HIV/AIDS in the ASEAN region.

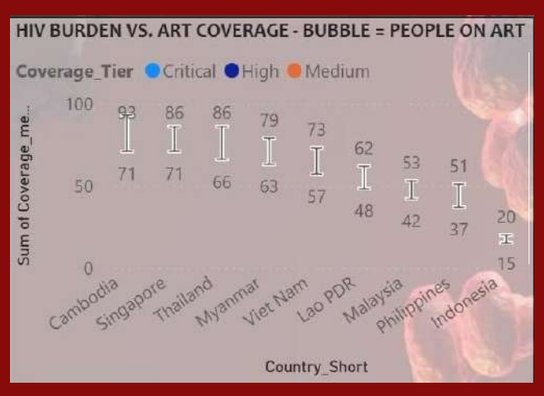
DATA ANALYSIS

SUM OF ESTIMATED MEDIAN OF PEOPLE LIVING WITH HIV BY COUNTRIES



EST PEOPLE LIVING WITH HIV BY MEDIAN

2M



METHODOLOGY

1. DATA COLLECTION

The dataset were collected from trusted sources including :

- WORLD HEALTH ORGANIZATION
- UNAIDS
- WORLD BANK
- DEPARTMENT OF STATISTICS MALAYSIA

2. DATA CLEANING

Data preprocessing was performed using Python to ensure data quality and consistency.

The cleaning process included:

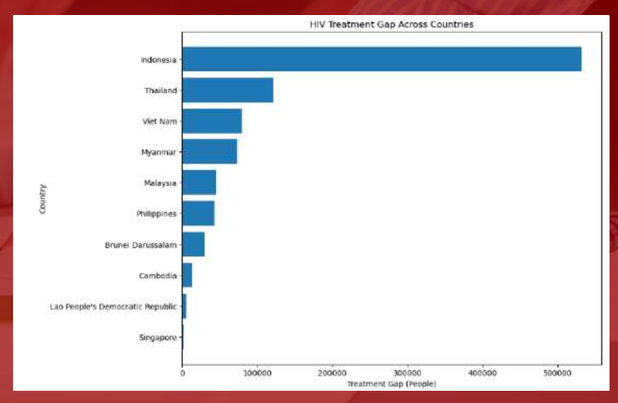
- converting variables from text format to numeric format
- replacing "Nodata" values with missing values
- extracting numerical estimates from range values
- handling missing values using median imputation
- verifying data types and removing inconsistencies

3. EXPLORATORY DATA ANALYSIS (EDA)

- Conducted using Power BI to understand the distribution of HIV burden and treatment coverage across countries.
- helps identify patterns and disparities in HIV/AIDS distribution within the region.

4. INEQUALITY ANALYSIS

TREATMENT GAP



This graph explain number of people living with HIV but not receiving ART.

Treatment to overcome HIV infection

ANTIRETROVIRAL THERAPY (ART)



- A combination of medications used to stopping HIV from reproducing.
- It's not a cure, but many people reach undetectable levels of HIV.

Type of drugs :

- Entry inhibitors.
- Capsid inhibitors.
- Nucleoside reverse transcriptase inhibitors (NRTIs).
- Non-nucleoside reverse transcriptase inhibitors (NNRTIs).
- Integrase inhibitors/integrase strand transfer inhibitors (INSTIs).
- Protease inhibitors.
- Pharmacokinetic enhancers.
- Combination medications.

Benefits / effects :



Keeps T-cell counts higher

Low risk of spreading HIV/AIDS

Immune system healthy



SOURCE : THE STAR NEWS

HIV/AIDS

HIV (HUMAN IMMUNODEFICIENCY VIRUS)

- A virus that attacks the immune system, especially the CD4 cells (T-helper cells) that help the body fight infections.

4 major routes/causes of transmission HIV :

unprotected vaginal or anal sexual intercourse

transmitted from a mother to her infant during pregnancy, childbirth and breastfeeding.

the sharing of contaminated needles, syringes, surgical equipment or other sharp instruments.

transfusion of contaminated blood

SOURCE : WORLD HEALTH ORGANISATION (WHO)

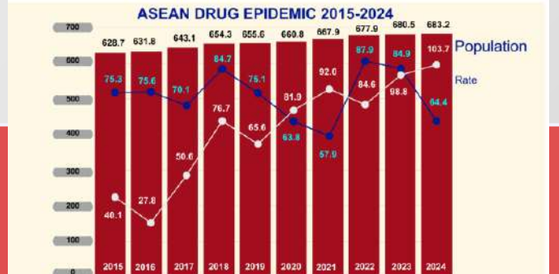
AIDS (ACQUIRED IMMUNODEFICIENCY SYNDROME)

- When HIV is not treated, it slowly damages the immune system until the body can no longer fight infections. At this stage, the person develops AIDS.

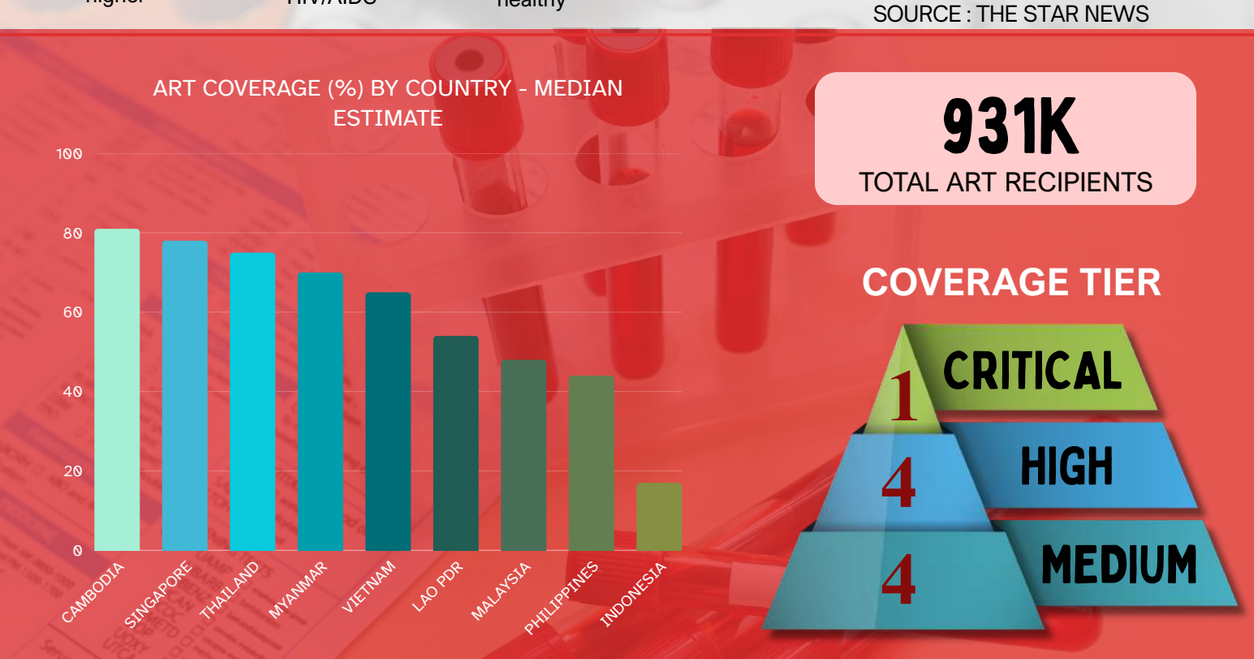
KEY POPULATIONS OF HIV/AIDS

- *Transgender*
- *Sex workers*
- *People in Prison and closed settings*
- *People with drug injection*
- *Gay men*

SOURCE : UNAIDS 2025 UPDATE



Source: ASEAN Narcotics Cooperation Center Reported on 2025



931K

TOTAL ART RECIPIENTS

COVERAGE TIER

CONCLUSION

- Inclusive and disaggregated data is essential to reveal hidden inequalities in HIV/AIDS prevalence and access to care across ASEAN countries.
- Antiretroviral therapy (ART) is essential for countering the infection of HIV towards ASEAN.
- Key populations must be controlled to avoid from increasing the impact of HIV/AIDS.
- ART is recommended because of the analysis shown the coverage influence the treatment gap.

REFERENCES

1. WORLD HEALTH ORGANISATION (WHO)
2. ASEAN NARCOTICS COOPERATION CENTER
3. UNAIDS
4. DEPARTMENT OF STATISTICS MALAYSIA
5. WORLD BANK
6. STAR NEWS

3 GOOD HEALTH AND WELL-BEING
4 QUALITY EDUCATION
8 DECENT WORK AND ECONOMIC GROWTH
12 RESPONSIBLE CONSUMPTION AND PRODUCTION

